

● HARD

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Circles

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02

ANSWER KEY · SAT QUESTION BANK

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Q1**A****A. -3**

Choice A is correct. A line that's tangent to a circle is perpendicular to the radius of the circle at the point of tangency. It's given that the circle has its center at $-4, -6$ and line k is tangent to the circle at the point $-7, -7$. The slope of a radius defined by the points q, r and s, t can be calculated as $\frac{t-r}{s-q}$. The points $-7, -7$ and $-4, -6$ define the radius of the circle at the point of tangency. Therefore, the slope of this radius can be calculated as $\frac{-6 - -7}{-4 - -7}$, or $\frac{1}{3}$. If a line and a radius are perpendicular, the slope of the line must be the negative reciprocal of the slope of the radius. The negative reciprocal of $\frac{1}{3}$ is -3 . Thus, the slope of line k is -3 .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the slope of the radius of the circle at the point of tangency, not the slope of line k .

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**C****C.** $\sqrt{167}$

Choice C is correct. It's given that in the xy -plane, the graph of the given equation is a circle. The equation of a circle in the xy -plane can be written in the form $x - h^2 + y - k^2 = r^2$, where h, k is the center of the circle and r is the length of the circle's radius. Subtracting $6y$ from both sides of the equation $x^2 + 14x + y^2 = 6y + 109$ yields $x^2 + 14x + y^2 - 6y = 109$. By completing the square, this equation can be rewritten as $x^2 + 14x + \frac{14^2}{2} + y^2 - 6y + \frac{-6^2}{2} = 109 + \frac{14^2}{2} + \frac{-6^2}{2}$. This equation can be rewritten as $x^2 + 14x + 49 + y^2 - 6y + 9 = 109 + 49 + 9$, or $x + 7^2 + y - 3^2 = 167$. Therefore, $r^2 = 167$. Taking the square root of both sides of this equation yields $r = \sqrt{167}$ and $r = -\sqrt{167}$. Since r is the length of the circle's radius, r must be positive. Therefore, the length of the circle's radius is $\sqrt{167}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3**B****B. 12π**

Choice B is correct. Because segments OA and OB are radii of the circle centered at point O, these segments have equal lengths. Therefore, triangle AOB is an isosceles triangle, where angles OAB and OBA are congruent base angles of the triangle. It's given that angle OAB measures 30° . Therefore, angle OBA also measures 30° . Let x° represent the measure of angle AOB. Since the sum of the measures of the three angles of any triangle is 180° , it follows that $30^\circ + 30^\circ + x^\circ = 180^\circ$, or $60^\circ + x^\circ = 180^\circ$. Subtracting 60° from both sides of this equation yields $x^\circ = 120^\circ$, or $\frac{2\pi}{3}$ radians. Therefore, the measure of angle AOB, and thus the measure of arc $\overline{AB}$, is $\frac{2\pi}{3}$ radians. Since $\overline{OC}$ is a radius of the given circle and its length is 18, the length of the radius of the circle is 18. Therefore, the length of arc $\overline{AB}$ can be calculated as $\left(\frac{2\pi}{3}\right)(18)$, or 12π .

Choices A, C, and D are incorrect and may result from conceptual or computational errors.

Q4**100**

The correct answer is 100. An equation of a circle in the xy -plane can be written as $(x - t)^2 + (y - u)^2 = r^2$, where the center of the circle is (t, u) , the radius of the circle is r , and where t , u , and r are constants. It's given that the equation of circle A is $(x + 5)^2 + (y - 5)^2 = 25$, which is equivalent to $(x + 5)^2 + (y - 5)^2 = 5^2$. Therefore, the center of circle A is $(-5, 5)$ and the radius of circle A is 5. It's given that circle B has the same center as circle A and that the radius of circle B is two times the radius of circle A. Therefore, the center of circle B is $(-5, 5)$ and the radius of circle B is $2(5)$, or 10. Substituting -5 for t , 5 for u , and 10 for r in the equation $(x - t)^2 + (y - u)^2 = r^2$ yields $(x + 5)^2 + (y - 5)^2 = 10^2$, which is equivalent to $(x + 5)^2 + (y - 5)^2 = 100$. It follows that the equation of circle B in the xy -plane is $(x + 5)^2 + (y - 5)^2 = 100$. It's also given that the equation defining circle B in the xy -plane is $(x + 5)^2 + (y - 5)^2 = k$, where k is a constant. Therefore, the value of k is 100.

Q5**10**

The correct answer is 10. It's given that the graph of $x^2 + x + y^2 + y = \frac{199}{2}$ in the xy -plane is a circle. The equation of a circle in the xy -plane can be written in the form $x - h^2 + y - k^2 = r^2$, where the coordinates of the center of the circle are h, k and the length of the radius of the circle is r . The term $x - h^2$ in this equation can be obtained by adding the square of half the coefficient of x to both sides of the given equation to complete the square. The coefficient of x is 1. Half the coefficient of x is $\frac{1}{2}$. The square of half the coefficient of x is $\frac{1}{4}$. Adding $\frac{1}{4}$ to each side of $x^2 + x + y^2 + y = \frac{199}{2}$ yields $x^2 + x + \frac{1}{4} + y^2 + y = \frac{199}{2} + \frac{1}{4}$, or $x + \frac{1}{2} + y^2 + y = \frac{199}{2} + \frac{1}{4}$. Similarly, the term $y - k^2$ can be obtained by adding the square of half the coefficient of y to both sides of this equation, which yields $x + \frac{1}{2} + y^2 + y + \frac{1}{4} = \frac{199}{2} + \frac{1}{4} + \frac{1}{4}$, or $x + \frac{1}{2} + y + \frac{1}{2} = \frac{199}{2} + \frac{1}{4} + \frac{1}{4}$. This equation is equivalent to $x + \frac{1}{2} + y + \frac{1}{2} = 100$, or $x + \frac{1}{2} + y + \frac{1}{2} = 10^2$. Therefore, the length of the circle's radius is 10.

Q6

C

C. 10, 2

Choice C is correct. It's given that the circle has its center at $-1, 1$ and that line t is tangent to this circle at the point $5, -4$. Therefore, the points $-1, 1$ and $5, -4$ are the endpoints of the radius of the circle at the point of tangency. The slope of a line or line segment that contains the points a, b and c, d can be calculated as $\frac{d-b}{c-a}$. Substituting $-1, 1$ for a, b and $5, -4$ for c, d in the expression $\frac{d-b}{c-a}$ yields $\frac{-4-1}{5-(-1)}$, or $-\frac{5}{6}$. Thus, the slope of this radius is $-\frac{5}{6}$. A line that's tangent to a circle is perpendicular to the radius of the circle at the point of tangency. It follows that line t is perpendicular to the radius at the point $5, -4$, so the slope of line t is the negative reciprocal of the slope of this radius. The negative reciprocal of $-\frac{5}{6}$ is $\frac{6}{5}$. Therefore, the slope of line t is $\frac{6}{5}$. Since the slope of line t is the same between any two points on line t , a point lies on line t if the slope of the line segment connecting the point and $5, -4$ is $\frac{6}{5}$. Substituting choice C, $10, 2$, for a, b and $5, -4$ for c, d in the expression $\frac{d-b}{c-a}$ yields $\frac{-4-2}{5-10}$, or $\frac{6}{5}$. Therefore, the point $10, 2$ lies on line t .

Choice A is incorrect. The slope of the line segment connecting $0, \frac{6}{5}$ and $5, -4$ is $\frac{-4-\frac{6}{5}}{5-0}$, or $-\frac{26}{25}$, not $\frac{6}{5}$.

Choice B is incorrect. The slope of the line segment connecting $4, 7$ and $5, -4$ is $\frac{-4-7}{5-4}$, or -11 , not $\frac{6}{5}$.

Choice D is incorrect. The slope of the line segment connecting $11, 1$ and $5, -4$ is $\frac{-4-1}{5-11}$, or $\frac{5}{6}$, not $\frac{6}{5}$.

Q7**-52**

The correct answer is -52. The equation of a circle in the xy -plane with its center at h, k and a radius of r can be written in the form $x - h^2 + y - k^2 = r^2$. It's given that a circle in the xy -plane has its center at $-5, 2$ and has a radius of 9. Substituting -5 for h , 2 for k , and 9 for r in the equation $x - h^2 + y - k^2 = r^2$ yields $x - 5^2 + y - 2^2 = 9^2$, or $x + 5^2 + y - 2^2 = 81$. It's also given that an equation of this circle is $x^2 + y^2 + ax + by + c = 0$, where a , b , and c are constants.

Therefore, $x + 5^2 + y - 2^2 = 81$ can be rewritten in the form $x^2 + y^2 + ax + by + c = 0$. The equation $x + 5^2 + y - 2^2 = 81$, or $x + 5x + 5 + y - 2y - 2 = 81$, can be rewritten as $x^2 + 5x + 5x + 25 + y^2 - 2y - 2y + 4 = 81$. Combining like terms on the left-hand side of this equation yields $x^2 + y^2 + 10x - 4y + 29 = 81$. Subtracting 81 from both sides of this equation yields $x^2 + y^2 + 10x - 4y - 52 = 0$, which is equivalent to $x^2 + y^2 + 10x + -4y + -52 = 0$. This equation is in the form $x^2 + y^2 + ax + by + c = 0$. Therefore, the value of c is -52.

Q8**A****A. $(2, -5)$**

Choice A is correct. The standard form for the equation of a circle is $(x-h)^2 + (y-k)^2 = r^2$, where (h,k) are the coordinates of the center and r is the length of the radius. According to the given equation, the center of the circle is $(6, -5)$. Let (x_1, y_1) represent the coordinates of point Q. Since point P $(10, -5)$ and point Q (x_1, y_1) are the endpoints of a diameter of the circle, the center $(6, -5)$ lies on the diameter, halfway between P and Q. Therefore, the following relationships hold: $\frac{x_1 + 10}{2} = 6$ and $\frac{y_1 + (-5)}{2} = -5$. Solving the equations for x_1 and y_1 , respectively, yields $x_1 = 2$ and $y_1 = -5$. Therefore, the coordinates of point Q are $(2, -5)$.

Alternate approach: Since point P $(10, -5)$ on the circle and the center of the circle $(6, -5)$ have the same y-coordinate, it follows that the radius of the circle is $10 - 6 = 4$. In addition, the opposite end of the diameter $\overline{PQ}$ must have the same y-coordinate as P and be 4 units away from the center. Hence, the coordinates of point Q must be $(2, -5)$.

Choices B and D are incorrect because the points given in these choices lie on a diameter that is perpendicular to the diameter $\overline{PQ}$. If either of these points were point Q, then $\overline{PQ}$ would not be the diameter of the circle. Choice C is incorrect because $(6, -5)$ is the center of the circle and does not lie on the circle.

Q9**D**

D. $x^2 + y + 1^2 = 49$

Choice D is correct. The graph in the xy -plane of an equation of the form $x - h^2 + y - k^2 = r^2$ is a circle with center h, k and a radius of length r . It's given that circle A is represented by $x^2 + y - 1^2 = 49$, which can be rewritten as $x^2 + y - 1^2 = 7^2$. Therefore, circle A has center $0, 1$ and a radius of length 7. Shifting circle A down two units is a rigid vertical translation of circle A that does not change its size or shape. Since circle B is obtained by shifting circle A down two units, it follows that circle B has the same radius as circle A, and for each point x, y on circle A, the point $x, y - 2$ lies on circle B. Moreover, if h, k is the center of circle A, then $h, k - 2$ is the center of circle B. Therefore, circle B has a radius of 7 and the center of circle B is $0, 1 - 2$, or $0, -1$. Thus, circle B can be represented by the equation $x^2 + y + 1^2 = 7^2$, or $x^2 + y + 1^2 = 49$.

Choice A is incorrect. This is the equation of a circle obtained by shifting circle A right 2 units.

Choice B is incorrect. This is the equation of a circle obtained by shifting circle A up 2 units.

Choice C is incorrect. This is the equation of a circle obtained by shifting circle A left 2 units.

Q10**B**

B. 13

Choice B is correct. Since it's given that P is the center of a circle with a radius of 9 inches, and that points Q and R lie on that circle, it follows that $\overline{PQ}$ and $\overline{RP}$ of triangle PQR each have a length of 9 inches. Let the length of $\overline{QR}$ be x inches. It follows that the perimeter of triangle PQR is $9 + 9 + x$ inches. Since it's given that the perimeter of triangle PQR is 31 inches, it follows that $9 + 9 + x = 31$, or $18 + x = 31$. Subtracting 18 from both sides of this equation gives $x = 13$. Therefore, the length, in inches, of $\overline{QR}$ is 13.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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